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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

ME2242-1- POWER PLANT ENGINEERING

(For MEC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Base load plants usually are:
A) microhydro
B) Nuclear / Thermal
C) Diesel
D) Gas turbine
2. Which of the following types of coal has the highest carbon content?
A) Lignite
B) Anthracite
C) Bituminous
D) Peat
3. The equipment used to dry and grind coal into fine particles is called
A) Stoker
B) Condenser
C) Pulveriser
D) Separator
4. Which ash handling system is most suitable when ash has to be conveyed over long distances?
A) Hydraulic system
B) Mechanical system
C) Pneumatic system
D) Manual system
5. The Benson boiler is also called
A) Once-through boiler
B) Fire-tube boiler
C) Drum type boiler
D) Shell boiler
6. The main function of the cooling tower in a thermal power plant is to
A) Increase the temperature of water
B) Cool the water after condensation
C) Increase the efficiency of the boiler
D) Remove the pollutants from the exhaust gases
7. Which of the following is not a boiler accessory?
A) Economizer
B) Air preheater
C) Superheater
D) Condenser
8. The device used at the top of cooling tower to prevent water droplets from escaping is called
A) Drift eliminator
B) Spray nozzle
C) Demister
D) Water distributor
9. The commonly used material for making solar cells is
A) Copper
B) Silicon
C) Aluminum
D) Lead
10. The power available in wind is proportional to
A) Wind velocity (V)
B) V^3
C) V^2
D) V^4
11. Location of a surge tank, in a hydroelectric power station, is near to
A) Turbine
B) Reservoir
C) Tail race
D) Dam
12. The function of the draft tube in a hydro turbine is
A) Carry water from reservoir
B) Store water temporarily
C) Discharge water to tailrace and recover pressure head
D) Increase water velocity



13. A gas turbine works on which cycle?
 - A) Carnot
 - B) Dual cycle
 - C) Brayton
 - D) Rankine cycle
14. In an open cycle gas turbine plant, the working fluid after expansion is
 - A) Re-circulated to compressor
 - B) Discharged to atmosphere
 - C) Sent to condenser
 - D) Mixed with feed water
15. The function of control rods in a nuclear reactor is to
 - A) Transfer heat to coolant
 - B) Absorb excess neutrons and control chain reaction
 - C) Slow down neutrons
 - D) Increase moderator efficiency
16. In a Pressurized Water Reactor (PWR), the moderator used is
 - A) Heavy water
 - B) Ordinary light water
 - C) Graphite
 - D) Helium
17. Load factor of a power plant is defined as
 - A) Ratio of average load to maximum demand
 - B) Ratio of plant capacity to average load
 - C) Ratio of maximum load to average load
 - D) Ratio of energy generated to maximum possible energy
18. A high load factor means
 - A) Low average load
 - B) Lower cost per unit of generation
 - C) Plant is under-utilized
 - D) Higher fuel consumption
19. The tariff system that encourages consumers to use more electricity at higher load factor
 - A) Flat rate tariff
 - B) Two-part tariff
 - C) Block rate tariff
 - D) Maximum demand tariff
20. The site of a wind power plant is selected mainly based on
 - A) Water availability
 - B) Soil fertility
 - C) Average wind velocity and pattern
 - D) Distance from coal mines

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

	Marks	BT*	CO*	PO*
1. a) Explain the operation of a spreader stoker with the help of neat sketch.	8	L*2	1	1
b) What is the function of a steam generator? Explain La'Mont boiler with neat sketch.	8	L2	2	1
2. a) Explain unit and bin system of pulverized coal handling. Mention the advantages and disadvantages of each system.	8	L2	1	1
b) What is draught? Explain Natural draught system with sketch and mention the advantages.	8	L2	2	1
3. a) List any four advantages of pulverized coal firing. Explain cyclone furnace with neat sketch.	8	L2	1	1
b) Explain the function and benefits of the following i) Economizer ii) Desuperheater iii) Air preheater iv) Reheater	8	L2	2	1

Unit – II

4. a) The run off data of a river at a particular site is tabulated in Table 1

Month	Mean discharge (millions of cu.m)	Month	Mean discharge (millions of cu.m)
January	30	July	80
February	25	August	100
March	20	September	110
April	0	October	65
May	10	November	45
June	50	December	30

- i) Draw the hydrograph and find mean flow
 ii) Draw the flow duration curve
 iii) Find the power developed if the head available is 90 m and the overall efficiency of generation is 86%. Assume each month of 30 days.
- 8 L3 3 1
- b) Mention the difference between chemical and nuclear reactions. Also explain the CANDU –type reactor with neat sketch.
- 8 L2 4 1
5. a) Explain the function of the following in hydroelectric plant.
- i) Surge tank
 ii) Draft tube
- 8 L2 3 1
- b) Discuss the following with reference to gas turbine power plant
- i) Intercooling
 ii) Reheating
- 8 L2 4 1
6. a) Explain flat plate solar energy collector with neat sketch. Also mention the comparison between flat plate collector and concentrating collectors.
- 8 L2 3 1
- b) Explain functions of the following in nuclear power plant
- i) Moderator
 ii) Reflector
 iii) Coolant
 iv) Cladding
- 8 L2 4 1
- Unit – III**
7. a) Explain what you understand by base load and peaking load. Why are base load plants loaded heavily?
- 8 L2 5 1
- b) A thermal power plant consists of two 60 MW units, each running for 8000 hours, and one 30 MW unit running for 2000 hours per year. The energy produced by the plant is 876×10^6 kWh per year. Determine the plant load factor and plant use factor. Consider the maximum load as equal to the plant capacity.
- 8 L3 5 1
8. a) Discuss the factors which should be considered while selecting a site for a hydroelectric plant.
- 8 L2 5 1
- b) Calculate the cost of generation per kwh for a power station having the following data:
 Installed capacity of the plant = 200 MW
 Capital cost = Rs. 400 crores
 Rate of interest and depreciation = 12%
 Annual cost of fuel, salaries and taxation = Rs. 5 crores.
 Load factor = 50%
 Also estimate the saving in cost per kwh if the annual load factor is raised to 60%.
- 8 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
